

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

MAY/JUNE EXAMINATION

BACS2163 SOFTWARE ENGINEERING

WEDNESDAY, 28 MAY 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR IN DATA SCIENCE (HONOURS)

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) TKM software development company will develop an E-learning Management System for a university. The system should allow students and educators to search for the class schedule, learning materials, assignments and to generate attendance reports. The system will be developed by implementing the **FOUR (4)** layers of software engineering technology. Describe each layer based on the system. (12 marks)
- b) A software development company is working on a high-priority project with tight deadlines. However, the development team faces declining motivation, impacting productivity and innovation. To address these, the project manager applies Maslow's Hierarchy of Needs to improve team engagement and performance. Identify and describe any **THREE (3)** levels of needs in Maslow's motivation model which the project manager can apply. (9 marks)
- c) A financial software company developed a Banking Management System (BMS) which is used by multiple banks for transactions, reporting, and customer management. While the system is stable and functional, customer feedback highlights the need for improved performance, a modern UI, and new features like AI-driven fraud detection. To address these needs, the company decides to implement an appropriate maintenance. Suggest the type of maintenance and provide any **ONE (1)** reason. (2 + 2 marks)

[Total: 25 marks]

Question 2

- a) Government agency is developing an E-Government Portal to provide online services such as tax filing, business registration, and public record access. Since multiple stakeholders have different perspectives on the system requirements, the development team adopts the viewpoint hierarchy model to ensure a structured approach.
- (i) Identify any **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in requirements gathering and analysis. (4 marks)
 - (ii) Identify any **SIX (6)** viewpoints of E-Government Portal. Group them into categories and construct a viewpoint hierarchy. (12 marks)
- b) A Real-Time Operating System (RTOS) is a specialised operating system designed to manage tasks that must be executed swiftly and within strict time constraints by implementing some key attributes. Discuss any **THREE (3)** attributes. (9 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

- a) A Fintech company is developing a mobile banking application with intuitive, accessible, and user-friendly Graphical User Interfaces (GUI). Discuss the **SIX (6)** user interface design principles the company must incorporate in the application. (12 marks)
- b) A global retail company deploys a Customer Relationship Management (CRM) System with headquarters in New York controlling global customer policies. Regional offices in London, Tokyo, and Sydney manage localised marketing campaigns, customer support, and sales strategies. The system integrates centralised customer data with region-specific preferences, ensuring a consistent but flexible customer experience worldwide. Recommend and illustrate a suitable system organisation model for the proposed project, providing **THREE (3)** reasons to justify your recommendation. (13 marks)

[Total: 25 marks]

Question 4

- a) An online retailer is developing a new E-Commerce Platform to handle product listings, customer orders, payments, and logistics. Since multiple modules interact within the system, integration testing is essential to ensure seamless communication between different components. Even though the payment processing module are still under development, testing can be started with high-level features. Analyse which integration testing technique would suit this project and provide **THREE (3)** reasons to justify your answer. (1 + 6 marks)
- b) A university intends to develop an Online Student Enrolment System to facilitate course registration, fee payments, and student record management. The main challenges in this project are the requirements continuously change even last minute, and the stakeholders would like to conduct product progress reviews every two weeks. Propose an appropriate software process model and justify your recommendation with **THREE (3)** reasons. (2 + 6 marks)
- c) Conducting a feasibility study is a crucial step in any project, as it helps identify potential risks, costs, and technical challenges before development begins. Analyse any **FIVE (5)** feasibility studies that can be used in a project. (10 marks)

[Total: 25 marks]